ANALYSTLAB AFRICA

Machine Learning Internship Programme

Data Preprocessing Report

Week 2: Data Preprocessing & Feature Engineering

Project: ShopSmart Retail Analytics

Field	Value
Programme	Machine Learning Internship
Week	Week 2
Document	Data Preprocessing Report
Dataset	ShopSmart Retail (multi-table → customer-level ML matrix)
Final artefact	ShopSmart_ML_Ready_Customers.csv

1. Executive Summary

This report documents the complete preprocessing pipeline applied to the ShopSmart retail data. Starting from nine source tables, the pipeline produces a clean, encoded, scaled and feature-selected customer-level dataset ready for machine learning. All decisions are recorded below so that the work is reproducible and auditable.

2. Data Quality Issues Identified

Inspection of the raw tables produced the following findings:

Issue	Observation	Impact if untreated
Missing values	None in core tables (Customers, Orders, Products, Stores, Returns)	Low – no imputation required on base tables
Duplicates	No full-row or primary-key duplicates	None
Date formats	Stored as strings (mm/dd/yyyy)	Cannot compute tenure or time features
Discount semantics	Values 0–20 → percentage, not absolute amount	Order totals would be systematically wrong
Negative tenure	Some Order_Date < Registration_Date in source data	Invalid tenure feature
Gender noise	Occasional label inconsistency vs names	Minor noise in one categorical
High cardinality	City has many unique values	Sparse one-hot explosion if encoded
Return rate	≈ 5% of orders returned	Class imbalance for return models
Outliers	Present in total_spend, avg_order_value, total_orders	Distort scaling and linear models

Table 1: Data quality findings

3. Data Cleaning Decisions

Step	Action	Justification
Duplicates	drop_duplicates() on all tables	Defensive hygiene
Dates	pd.to_datetime(..., format='%m/%d/%Y')	Enable tenure and calendar features
Strings	Strip whitespace on IDs and labels	Prevent silent join / groupby mismatches
Numerics	pd.to_numeric(..., errors='coerce')	Guard against bad parses
Order_Total	Qty × Price × (1 – Discount/100) + Shipping	Discount is a percentage
Returns summary	Explicit groupby on Order_ID	Reproducible merge (fixes Week 1 gap)
Return nulls	Fill count/amount with 0; reason with 'Not Returned'	Orders never returned are valid zeros
Inventory nulls	Fill with column median	Acceptable first-pass strategy
Tenure	Clip negative values to 0	Order before registration is data noise
PII / City	Drop names, email, phone, City	Privacy and dimensionality control

Table 2: Cleaning decisions

4. Feature Engineering

An order-level master table was first constructed (Orders enriched with Customers, Products, Stores, returns summary, employee summary and inventory summary). Behavioural features were then aggregated to customer level — the primary grain for segment, CLV and propensity models.

Feature	Definition	Business rationale
total_orders	Count of distinct orders	RFM – Frequency
total_spend	Sum of Order_Total	RFM – Monetary
avg_order_value	Mean Order_Total	Basket size
avg_discount	Mean discount %	Promotion sensitivity
avg_shipping	Mean shipping cost	Cost-to-serve indicator
days_as_customer	Last order – registration (clipped ≥ 0)	Tenure / lifecycle stage
unique_products	Distinct products purchased	Exploration / diversity
unique_stores	Distinct stores used	Multi-store behaviour
pct_online	Share of orders via Online channel	Channel preference
pct_completed	Share of Completed orders	Fulfilment success
return_count / rate	Sum and rate of returned orders	Return propensity
preferred_payment	Modal payment method	Payment behaviour
most_bought_category	Modal product category	Category affinity

Table 3: Engineered customer-level features

5. Encoding & Scaling

Encoding

• **Ordinal encoding** for Customer_Segment (Bronze < Silver < Gold) — preserves natural order.
• **One-hot encoding** (drop_first=True) for Gender, State, preferred_payment, most_bought_category — avoids imposing false ordinality on nominal variables.
• City was dropped (high cardinality; State already captures geography).

Scaling

• **RobustScaler** for counts and monetary features (Age, total_orders, total_spend, avg_order_value, etc.) — uses median and IQR, therefore resistant to remaining outliers.
• **StandardScaler** for bounded ratio features (pct_online, pct_completed, return_rate).
Negative values after scaling are expected: they indicate observations below the centre (median/mean) and are valid inputs for downstream models.

6. Outlier Detection & Treatment

Outliers were detected on key numeric features using both the Interquartile Range rule (k = 1.5) and Z-score (|z| > 3). Extreme values on total_spend, avg_order_value and total_orders were **winsorized** at the 1st and 99th percentiles rather than deleted, so that every customer is retained while leverage on subsequent models is limited.

7. Feature Selection

1. **VarianceThreshold (0.01)** — removed near-constant columns that add no signal.
2. **Correlation filter (|r| > 0.85)** — dropped one feature from each highly collinear pair to reduce multicollinearity for linear models.

8. Final Machine-Learning-Ready Dataset

The output file **ShopSmart_ML_Ready_Customers.csv** contains one row per customer with:

- Customer_ID (identifier, not used as a model feature)
- Scaled numeric behavioural features
- Encoded categorical features (ordinal + one-hot)
- Customer_Segment_ord (0 = Bronze, 1 = Silver, 2 = Gold) as a ready target
- Human-readable Customer_Segment label retained for reporting

Null check on the final matrix is zero. The dataset is therefore ready for train/test split and model fitting.

9. Answers to the Week 2 Business Questions

#	Question	Answer
1	Data quality issues?	Core tables clean; date strings; % discount; negative tenure cases; minor Gender noise; ≈5% return rate.
2	Features needing preprocessing?	Dates, categoricals, monetary/count fields, high-cardinality City.
3	Encoding?	Ordinal for ordered Segment; One-Hot for nominal Gender, State, Payment, Category.
4	Scaling?	RobustScaler (counts/money) + StandardScaler (ratios).
5	Significant outliers?	Yes — spend and order counts; treated by winsorization (1%–99%).
6	Most useful features?	total_spend, total_orders, avg_order_value, return_rate, days_as_customer, pct_online, Segment.
7	Ready for training?	Yes — ShopSmart_ML_Ready_Customers.csv is complete and validated.

Table 4: Business question answers

10. Key Observations

• Building a customer-level feature matrix from an order-centric master is the right grain for segment, CLV and propensity work.
• Correcting the Order_Total formula (percentage discount) and explicitly defining the returns summary were essential reproducibility fixes relative to the Week 1 notebook.

• Winsorization is preferable to row deletion when every customer must remain in the modelling set.
• Scaled features legitimately contain negative values; these should not be confused with raw-data errors.
• Documenting every decision makes the pipeline suitable for handover and audit.

11. Limitations & Next Steps

Inventory features after median fill showed limited variation and may add little signal; a store- or category-level imputation could be explored later. Supplier data remains unused pending a usable product-level link. Future weeks can train models on the prepared matrix, address class imbalance for return prediction, and evaluate feature importance under cross-validation.